Assignment 16.4 Ai Assisted Coding

Htno:2303a51306

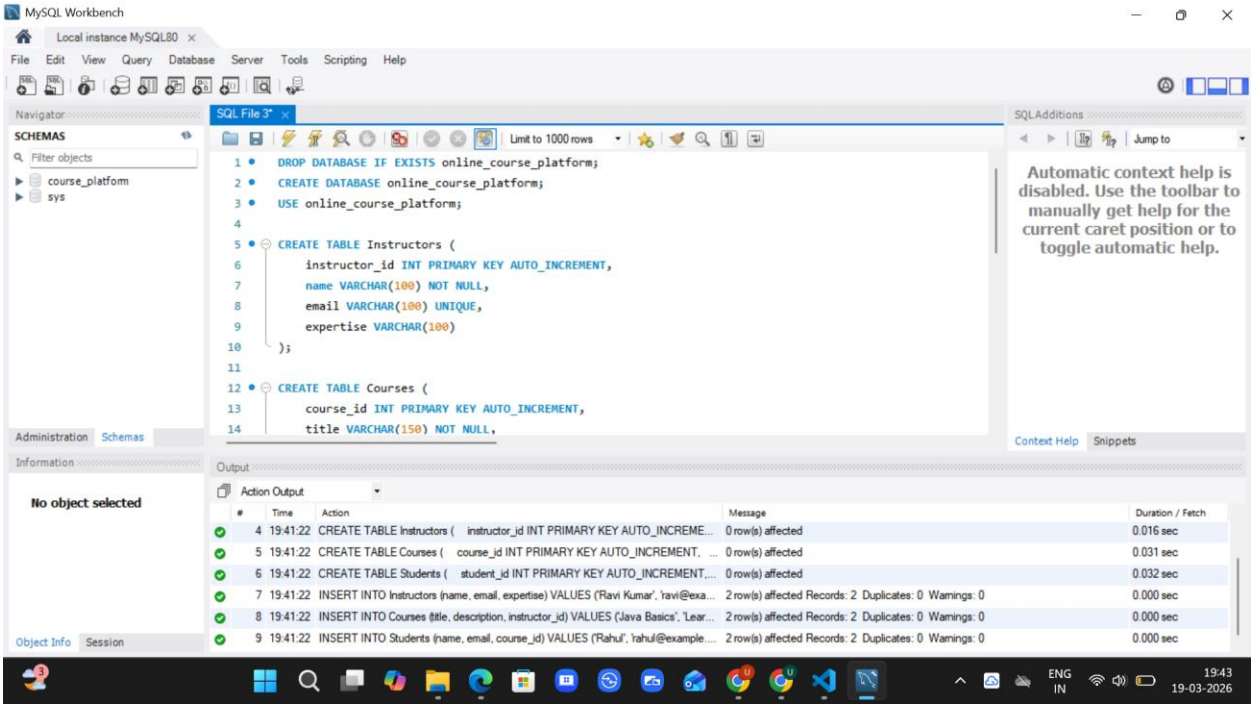
Btno:05

Task 1: Database Schema Design for an Online Course Platform

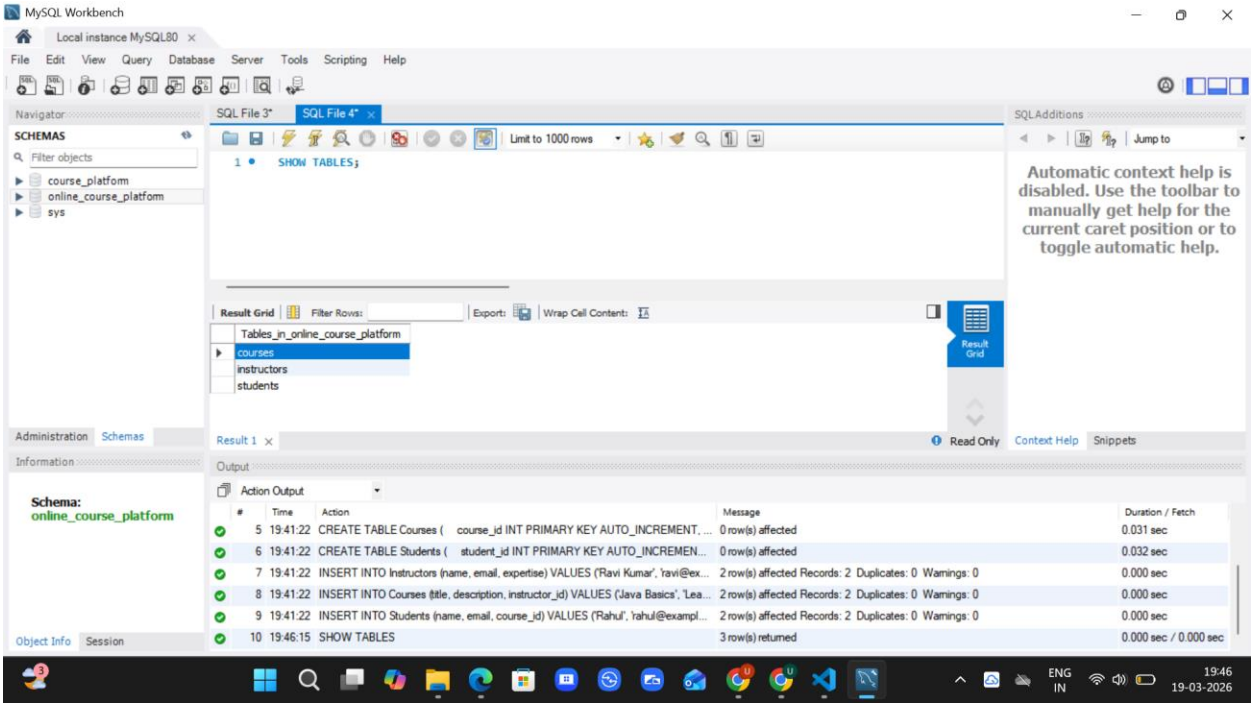
Prompt:

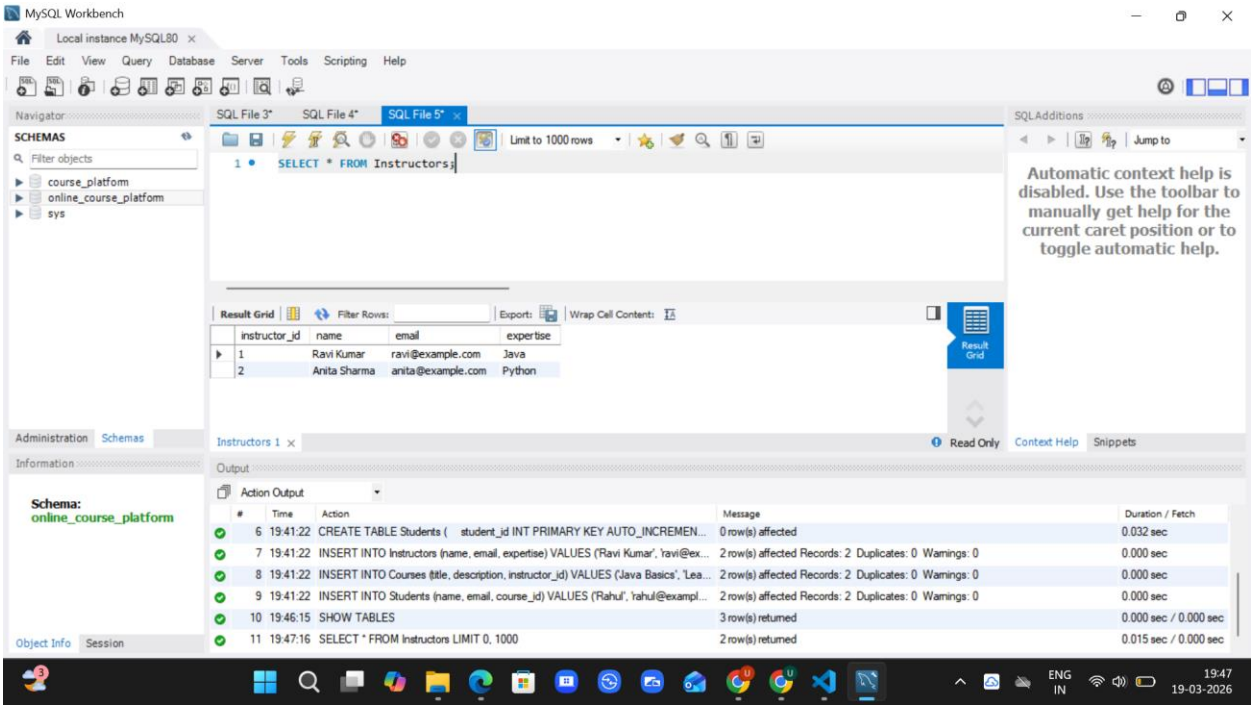
Generate SQL CREATE TABLE statements for an online course platform including Instructors, Courses, and Students. Ensure the use of primary keys, foreign key relationships, and appropriate data types. The schema should be normalized and logically structured.

Code:



Output:


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
course_platform
online_course_platform
sys
SQL File 3*
SQL File 4*
Limit to 1000 rows
1 SHOW TABLES;
SQL Additions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Tables_in_online_course_platform
courses
instructors
students
Result 1
Read Only
Context Help
Snippets
Administration
Schemas
Information
Schema:
online_course_platform
Output
Action Output
Time Action Message Duration / Fetch
5 19:41:22 CREATE TABLE Courses (course_id INT PRIMARY KEY AUTO_INCREMENT, ... 0 row(s) affected 0.031 sec
6 19:41:22 CREATE TABLE Students (student_id INT PRIMARY KEY AUTO_INCREMEN... 0 row(s) affected 0.032 sec
7 19:41:22 INSERT INTO Instructors (name, email, expertise) VALUES ('Ravi Kumar', 'ravi@ex... 2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0 0.000 sec
8 19:41:22 INSERT INTO Courses (title, description, instructor_id) VALUES ('Java Basics', 'Lea... 2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0 0.000 sec
9 19:41:22 INSERT INTO Students (name, email, course_id) VALUES ('Rahul', 'rahul@exampl... 2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0 0.000 sec
10 19:46:15 SHOW TABLES 3 row(s) returned 0.000 sec / 0.000 sec
Object Info
Session
ENG IN
19:46
19-03-2026


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
course_platform
online_course_platform
sys
SQL File 3*
SQL File 4*
SQL File 5*
Limit to 1000 rows
1 SELECT * FROM Instructors;
SQL Additions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
instructor_id name email expertise
1 Ravi Kumar ravi@example.com Java
2 Anita Sharma anita@example.com Python
Instructors 1
Read Only
Context Help
Snippets
Administration
Schemas
Information
Schema:
online_course_platform
Output
Action Output
Time Action Message Duration / Fetch
6 19:41:22 CREATE TABLE Students (student_id INT PRIMARY KEY AUTO_INCREMEN... 0 row(s) affected 0.032 sec
7 19:41:22 INSERT INTO Instructors (name, email, expertise) VALUES ('Ravi Kumar', 'ravi@ex... 2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0 0.000 sec
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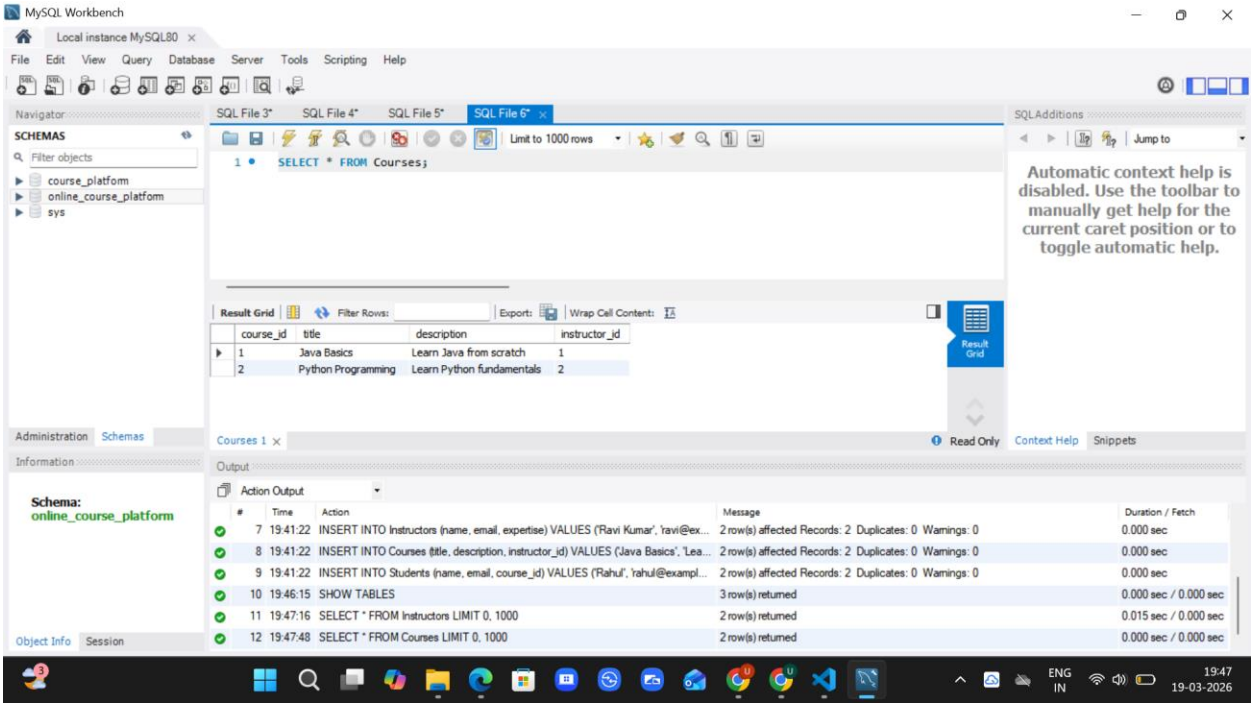
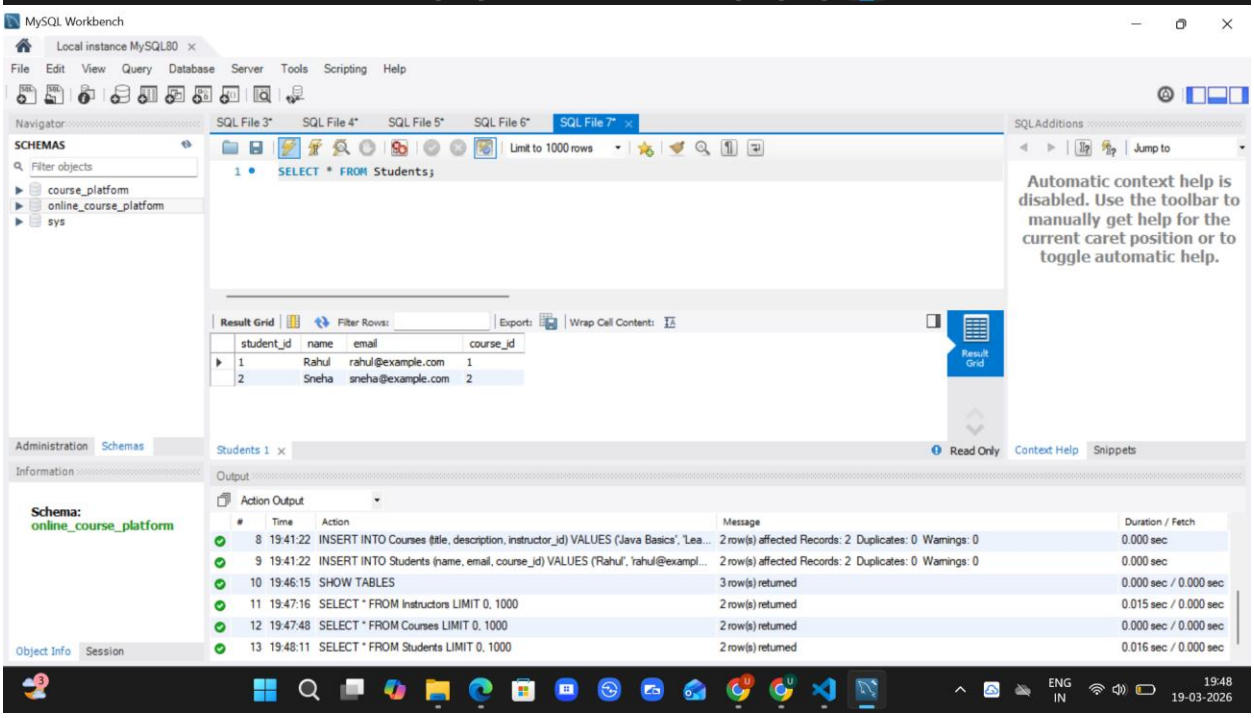
9 19:41:22 INSERT INTO Students (name, email, course_id) VALUES ('Rahul', 'rahul@exampl... 2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0 0.000 sec
10 19:46:15 SHOW TABLES 3 row(s) returned 0.000 sec / 0.000 sec
11 19:47:16 SELECT * FROM Instructors LIMIT 0, 1000 2 row(s) returned 0.015 sec / 0.000 sec
Object Info
Session
ENG IN
19:47
19-03-2026





Explanation:

-->Database Design

Created tables for Instructors, Courses, Students using CREATE TABLE with proper data types and primary keys.

-->Relationships (Foreign Keys)

Connected tables using foreign keys (e.g., course linked to instructor) to maintain data integrity.

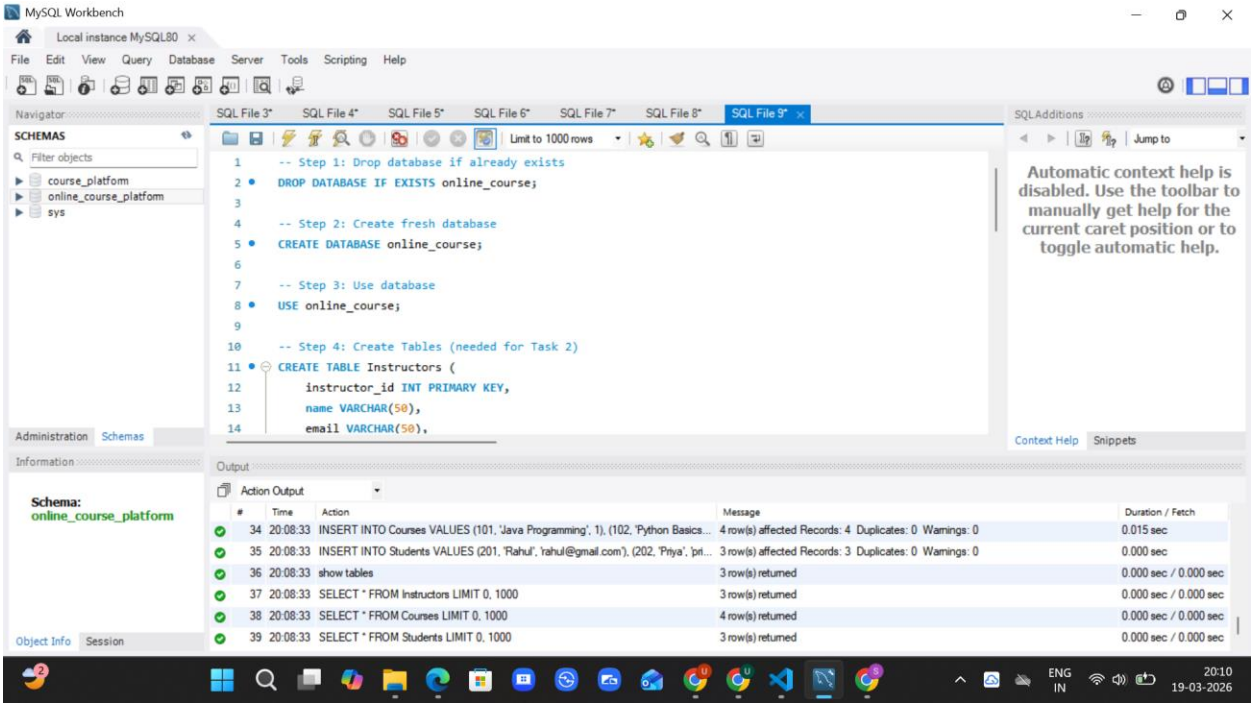
-->Normalization
Structured tables without duplication (separate tables for each entity) to follow basic normalization rules.

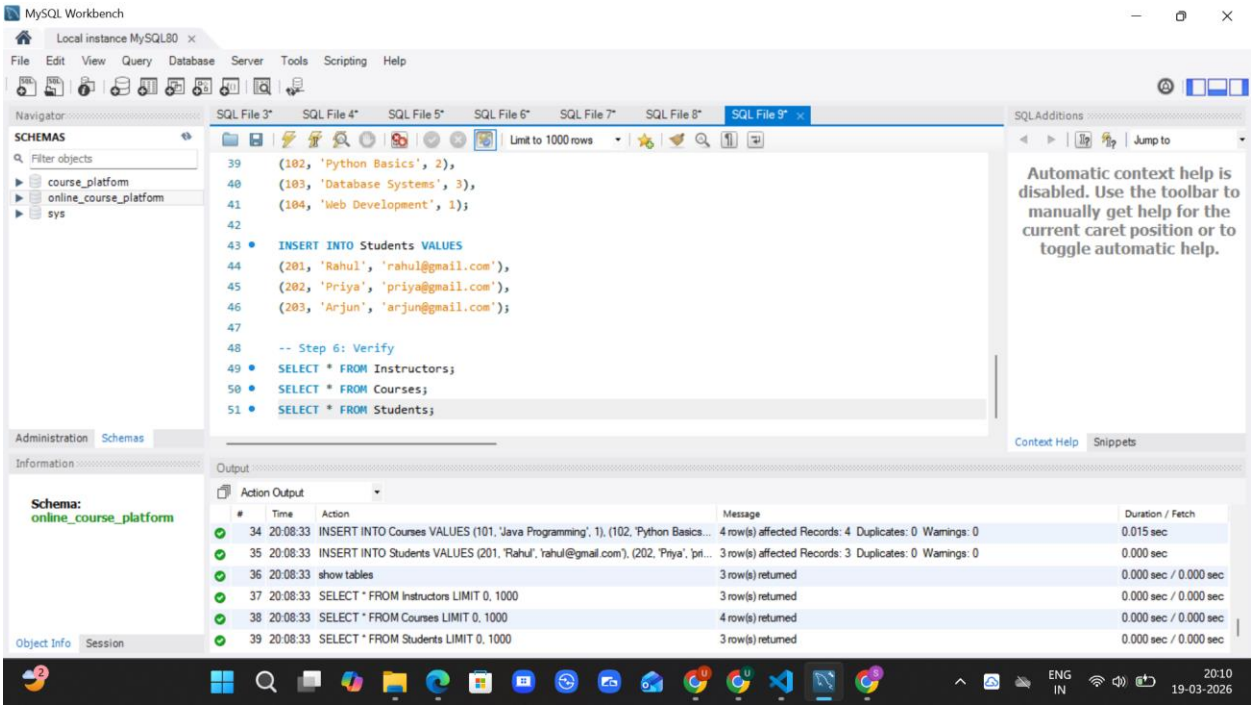
Task 2: Populate the Database with Sample Records

Prompt:

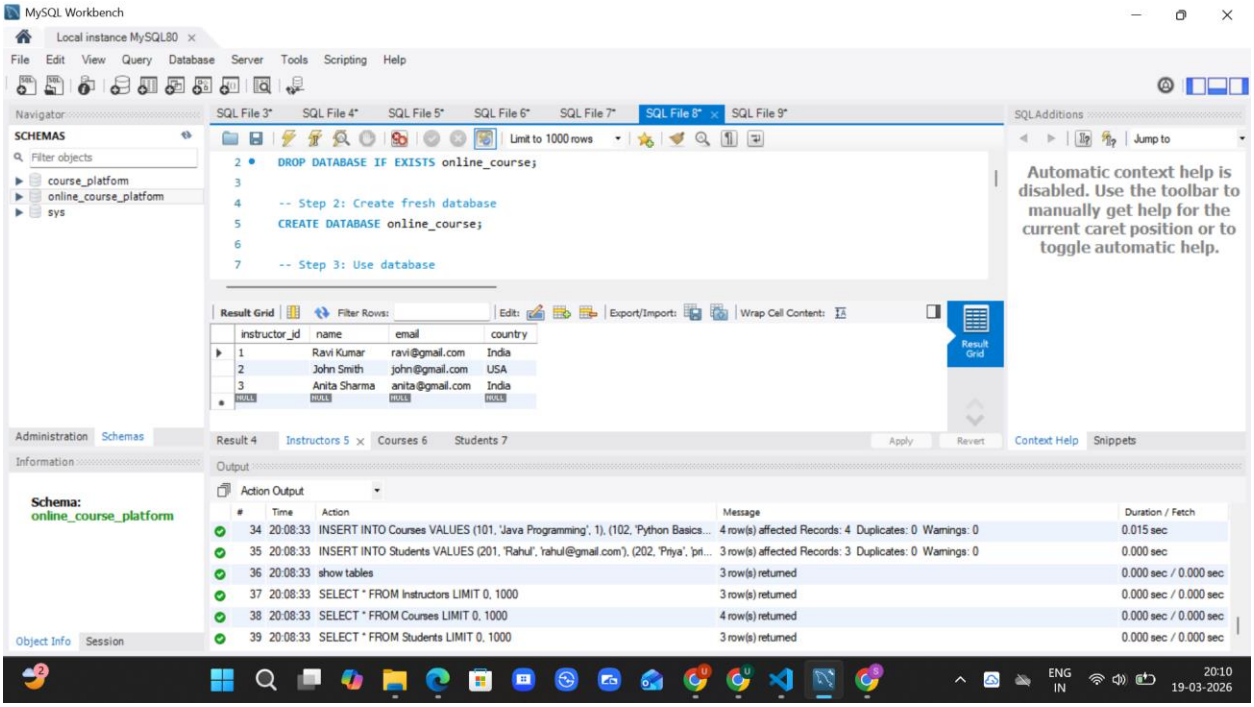
Generate SQL `INSERT` statements to populate the database with sample records including at least **3 instructors, 4 courses, and 3 students**, and verify the inserted data using `SELECT` queries.

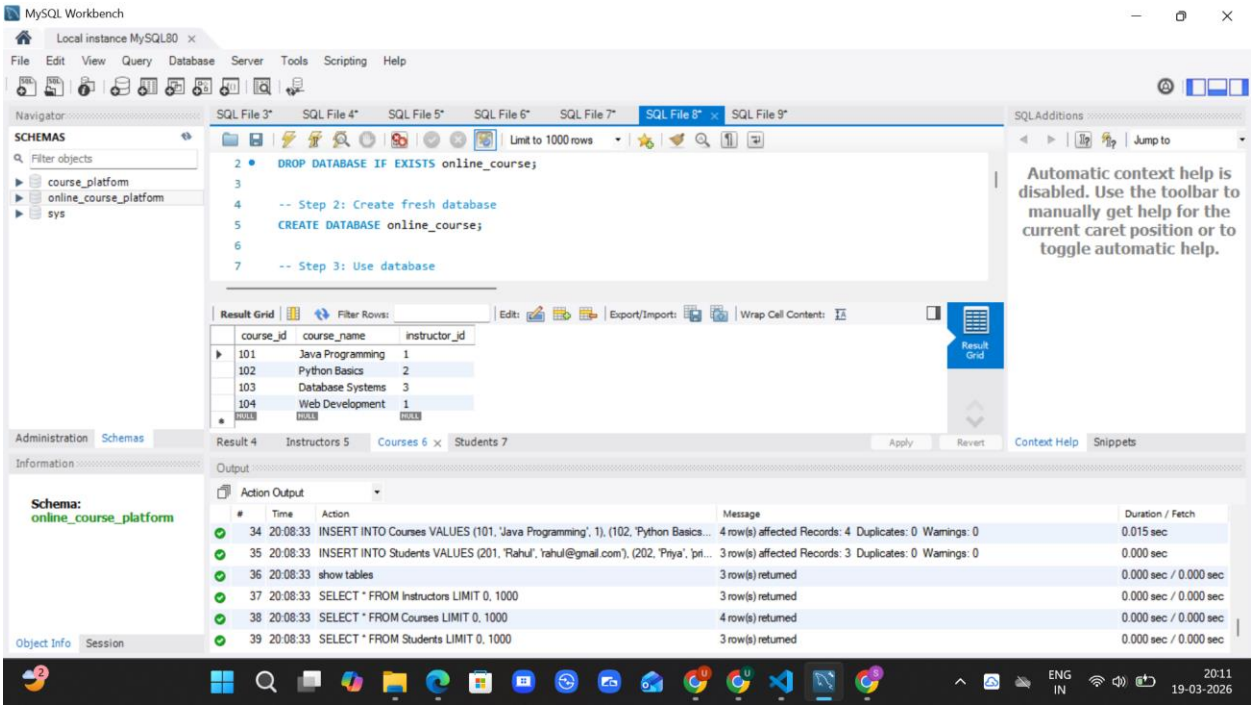
Code:

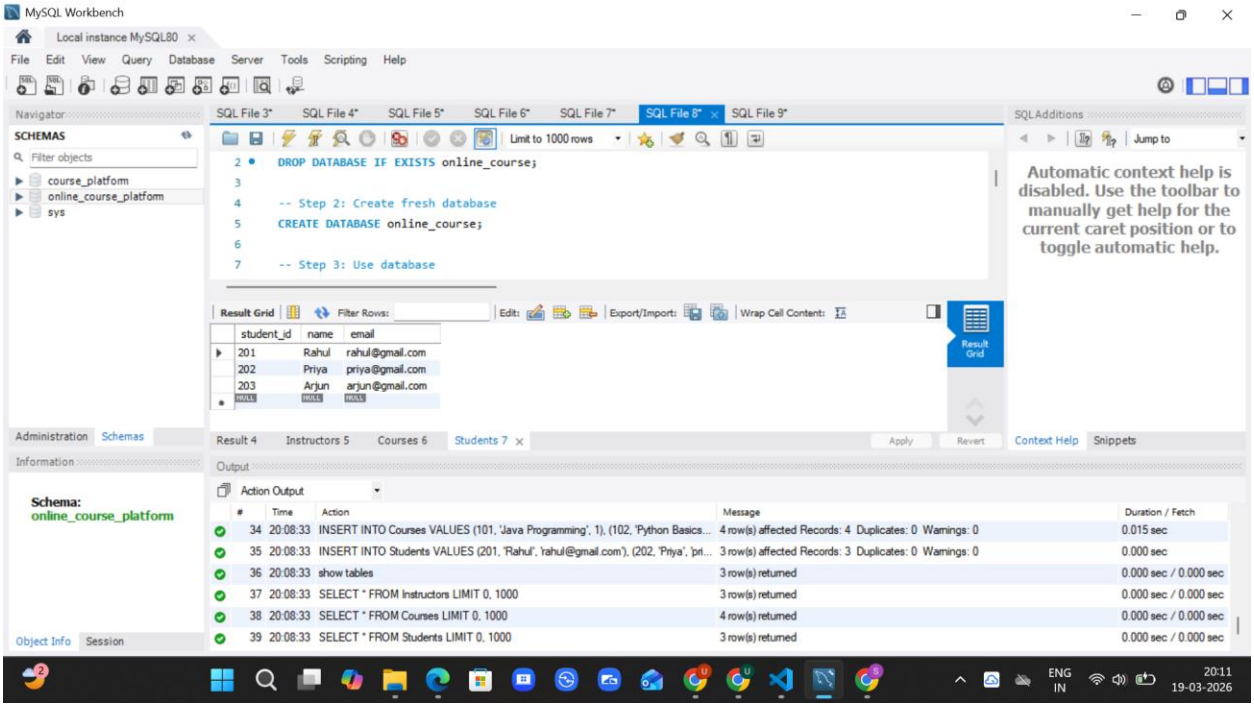

MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9*
Limit to 1000 rows
-- Step 1: Drop database if already exists
DROP DATABASE IF EXISTS online_course;
-- Step 2: Create fresh database
CREATE DATABASE online_course;
-- Step 3: Use database
USE online_course;
-- Step 4: Create Tables (needed for Task 2)
CREATE TABLE Instructors (
instructor_id INT PRIMARY KEY,
name VARCHAR(50),
email VARCHAR(50),
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Schema: online_course_platform
Output
Action Output
34 20:08:33 INSERT INTO Courses VALUES (101, 'Java Programming', 1), (102, 'Python Basics... 4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0 0.015 sec
35 20:08:33 INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'pri... 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 0.000 sec
36 20:08:33 show tables 3 row(s) returned 0.000 sec / 0.000 sec
37 20:08:33 SELECT * FROM Instructors LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
38 20:08:33 SELECT * FROM Courses LIMIT 0, 1000 4 row(s) returned 0.000 sec / 0.000 sec
39 20:08:33 SELECT * FROM Students LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9*
Limit to 1000 rows
(102, 'Python Basics', 2),
(103, 'Database Systems', 3),
(104, 'Web Development', 1);
INSERT INTO Students VALUES
(201, 'Rahul', 'rahul@gmail.com'),
(202, 'Priya', 'priya@gmail.com'),
(203, 'Arjun', 'arjun@gmail.com');
-- Step 6: Verify
SELECT * FROM Instructors;
SELECT * FROM Courses;
SELECT * FROM Students;
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Schema: online_course_platform
Output
Action Output
34 20:08:33 INSERT INTO Courses VALUES (101, 'Java Programming', 1), (102, 'Python Basics... 4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0 0.015 sec
35 20:08:33 INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'pri... 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 0.000 sec
36 20:08:33 show tables 3 row(s) returned 0.000 sec / 0.000 sec
37 20:08:33 SELECT * FROM Instructors LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
38 20:08:33 SELECT * FROM Courses LIMIT 0, 1000 4 row(s) returned 0.000 sec / 0.000 sec
39 20:08:33 SELECT * FROM Students LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec

Output:


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9*
Limit to 1000 rows
DROP DATABASE IF EXISTS online_course;
-- Step 2: Create fresh database
CREATE DATABASE online_course;
-- Step 3: Use database
Result Grid
instructor_id name email country
1 Ravi Kumar ravi@gmail.com India
2 John Smith john@gmail.com USA
3 Anita Sharma anita@gmail.com India
Schema: online_course_platform
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
34 20:08:33 INSERT INTO Courses VALUES (101, 'Java Programming', 1), (102, 'Python Basics... 4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0 0.015 sec
35 20:08:33 INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'pri... 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 0.000 sec
36 20:08:33 show tables 3 row(s) returned 0.000 sec / 0.000 sec
37 20:08:33 SELECT * FROM Instructors LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
38 20:08:33 SELECT * FROM Courses LIMIT 0, 1000 4 row(s) returned 0.000 sec / 0.000 sec
39 20:08:33 SELECT * FROM Students LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
20:10 19-03-2026


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9*
Limit to 1000 rows
DROP DATABASE IF EXISTS online_course;
-- Step 2: Create fresh database
CREATE DATABASE online_course;
-- Step 3: Use database
Result Grid
course_id course_name instructor_id
101 Java Programming 1
102 Python Basics 2
103 Database Systems 3
104 Web Development 1
Schema: online_course_platform
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
34 20:08:33 INSERT INTO Courses VALUES (101, 'Java Programming', 1), (102, 'Python Basics... 4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0 0.015 sec
35 20:08:33 INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'pri... 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 0.000 sec
36 20:08:33 show tables 3 row(s) returned 0.000 sec / 0.000 sec
37 20:08:33 SELECT * FROM Instructors LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
38 20:08:33 SELECT * FROM Courses LIMIT 0, 1000 4 row(s) returned 0.000 sec / 0.000 sec
39 20:08:33 SELECT * FROM Students LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
20:11 19-03-2026



Explanation:

-->Used DROP DATABASE to remove old data and start fresh

-->Recreated tables and inserted sample records

-->Verified data using SELECT queries

Task 3: Implement Core Database Operations (CRUD)

Prompt:

Perform SQL operations to **retrieve all courses, update a student's email, and delete a course**, then verify using `SELECT` queries.

Code:

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11*

Limit to 1000 rows

```sql
-- Step 1: Drop database to avoid already exists error
DROP DATABASE IF EXISTS online_course;

-- Step 2: Create database
CREATE DATABASE online_course;

-- Step 3: Use database
USE online_course;

-- Step 4: Create tables
CREATE TABLE Instructors (
    instructor_id INT PRIMARY KEY,
    name VARCHAR(50),
    email VARCHAR(50),
```

Navigator

SCHEMAS

Filter objects

- course_platform
- online_course_platform
- sys

Administration Schemas

Information

Schema:
online_course_platform

Object Info Session

SQL Additions

Jump to

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
48	20:13:42	INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'pri...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
49	20:13:42	SELECT * FROM Courses LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
50	20:13:42	UPDATE Students SET email = 'rahul_new@gmail.com' WHERE student_id = 201	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
51	20:13:42	DELETE FROM Courses WHERE course_id = 104	1 row(s) affected	0.000 sec
52	20:13:42	SELECT * FROM Students LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
53	20:13:42	SELECT * FROM Courses LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

ENG IN 20:14 19-03-2026

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11*

Limit to 1000 rows

```sql
-- Step 5: Insert sample data
INSERT INTO Instructors VALUES
(1, 'Ravi Kumar', 'ravi@gmail.com', 'India'),
(2, 'John Smith', 'john@gmail.com', 'USA'),
(3, 'Anita Sharma', 'anita@gmail.com', 'India');

INSERT INTO Courses VALUES
(101, 'Java Programming', 1),
(102, 'Python Basics', 2),
(103, 'Database Systems', 3),
(104, 'Web Development', 1);

INSERT INTO Students VALUES
(201, 'Rahul', 'rahul@gmail.com'),
```

Navigator

SCHEMAS

Filter objects

- course_platform
- online_course_platform
- sys

Administration Schemas

Information

Schema:
online_course_platform

Object Info Session

SQL Additions

Jump to

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

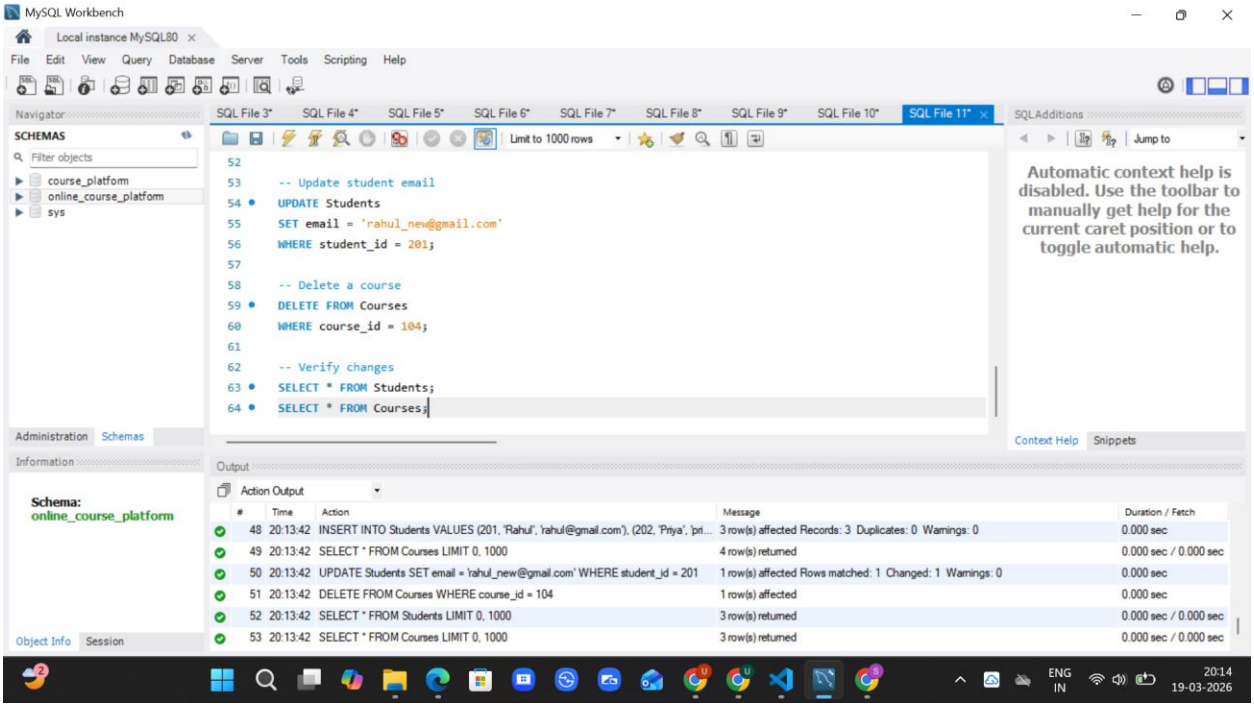
Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
48	20:13:42	INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'pri...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
49	20:13:42	SELECT * FROM Courses LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
50	20:13:42	UPDATE Students SET email = 'rahul_new@gmail.com' WHERE student_id = 201	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
51	20:13:42	DELETE FROM Courses WHERE course_id = 104	1 row(s) affected	0.000 sec
52	20:13:42	SELECT * FROM Students LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
53	20:13:42	SELECT * FROM Courses LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

ENG IN 20:14 19-03-2026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SCHEMAS
Filter objects
course_platform
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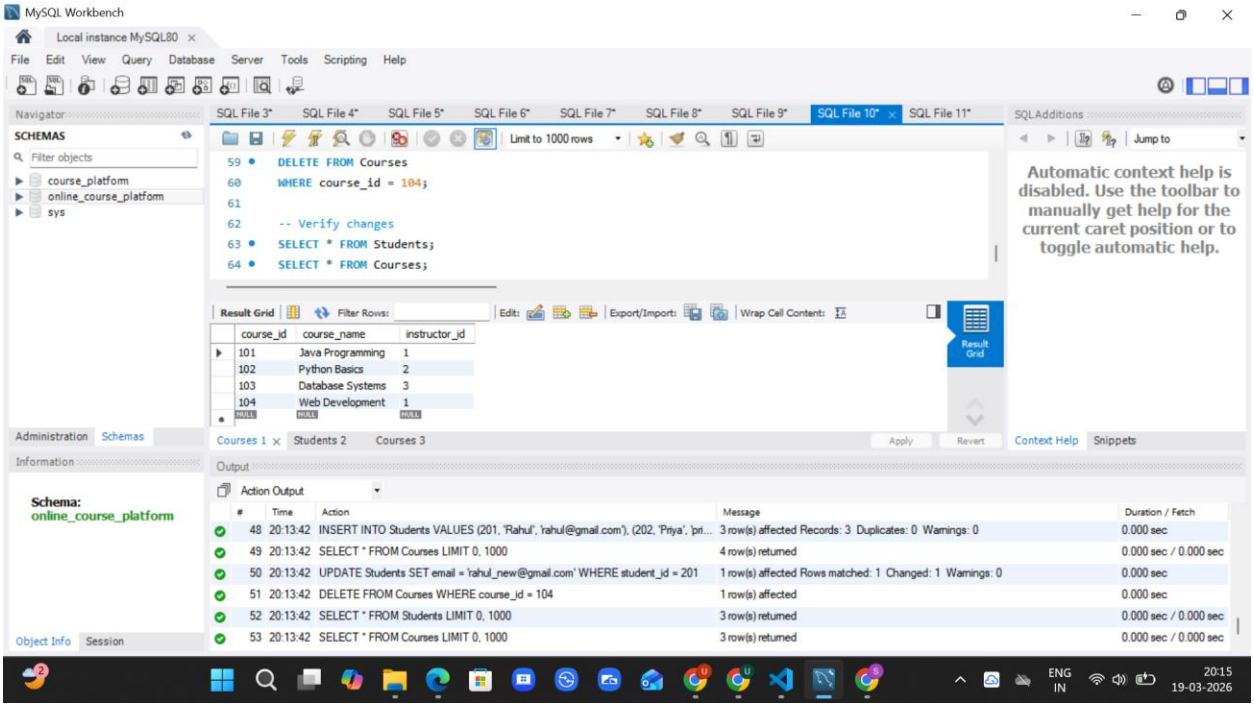
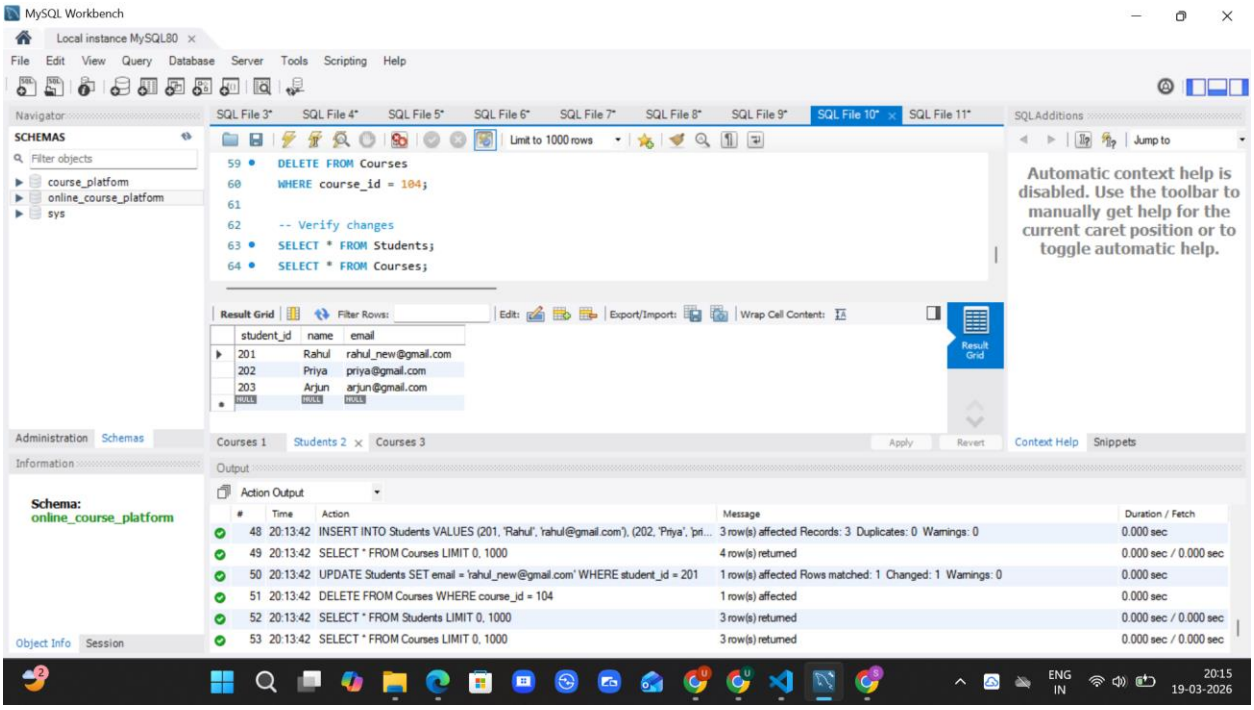
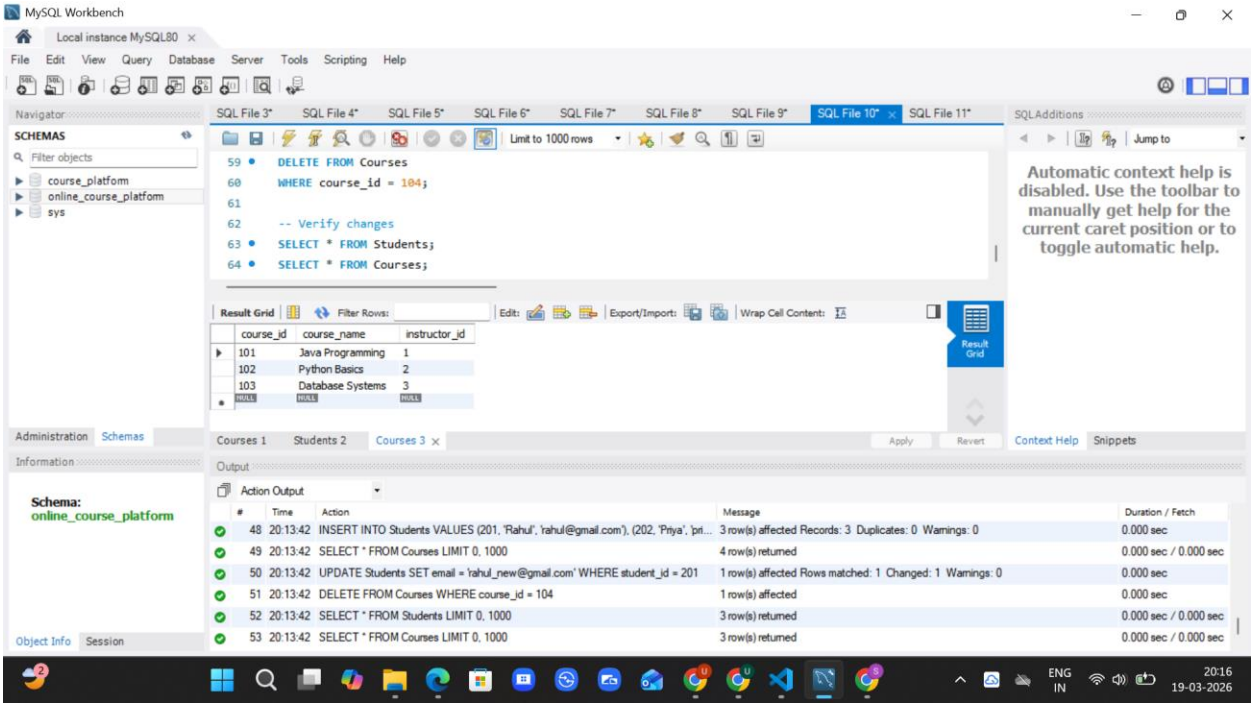
sys
Administration Schemas
Information
Schema:
online_course_platform
Object Info Session
SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11*
Limit to 1000 rows
52
53 -- Update student email
54 UPDATE Students
55 SET email = 'rahul_new@gmail.com'
56 WHERE student_id = 201;
57
58 -- Delete a course
59 DELETE FROM Courses
60 WHERE course_id = 104;
61
62 -- Verify changes
63 SELECT * FROM Students;
64 SELECT * FROM Courses;
SQL Additions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Output
Action Output
Time Action Message Duration / Fetch
48 20:13:42 INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'pri... 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 0.000 sec
49 20:13:42 SELECT * FROM Courses LIMIT 0, 1000 4 row(s) returned 0.000 sec / 0.000 sec
50 20:13:42 UPDATE Students SET email = 'rahul_new@gmail.com' WHERE student_id = 201 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 0.000 sec
51 20:13:42 DELETE FROM Courses WHERE course_id = 104 1 row(s) affected 0.000 sec
52 20:13:42 SELECT * FROM Students LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
53 20:13:42 SELECT * FROM Courses LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
ENG IN
20:14
19-03-2026

Output:


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11*
Limit to 1000 rows
59 DELETE FROM Courses
60 WHERE course_id = 104;
61
62 -- Verify changes
63 SELECT * FROM Students;
64 SELECT * FROM Courses;
Result Grid
course_id course_name instructor_id
101 Java Programming 1
102 Python Basics 2
103 Database Systems 3
104 Web Development 1
Schema: online_course_platform
Output
Action Output
48 20:13:42 INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'pri... 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 0.000 sec
49 20:13:42 SELECT * FROM Courses LIMIT 0, 1000 4 row(s) returned 0.000 sec / 0.000 sec
50 20:13:42 UPDATE Students SET email = 'rahul_new@gmail.com' WHERE student_id = 201 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 0.000 sec
51 20:13:42 DELETE FROM Courses WHERE course_id = 104 1 row(s) affected 0.000 sec
52 20:13:42 SELECT * FROM Students LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
53 20:13:42 SELECT * FROM Courses LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
ENG IN
20:15 19-03-2026


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11*
Limit to 1000 rows
59 DELETE FROM Courses
60 WHERE course_id = 104;
61
62 -- Verify changes
63 SELECT * FROM Students;
64 SELECT * FROM Courses;
Result Grid
student_id name email
201 Rahul rahul_new@gmail.com
202 Priya priya@gmail.com
203 Arjun arjun@gmail.com
Schema: online_course_platform
Output
Action Output
48 20:13:42 INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'pri... 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 0.000 sec
49 20:13:42 SELECT * FROM Courses LIMIT 0, 1000 4 row(s) returned 0.000 sec / 0.000 sec
50 20:13:42 UPDATE Students SET email = 'rahul_new@gmail.com' WHERE student_id = 201 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 0.000 sec
51 20:13:42 DELETE FROM Courses WHERE course_id = 104 1 row(s) affected 0.000 sec
52 20:13:42 SELECT * FROM Students LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
53 20:13:42 SELECT * FROM Courses LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
ENG IN
20:15 19-03-2026



Explanation:

-->Used SELECT to retrieve all course records

-->Updated student email using UPDATE statement

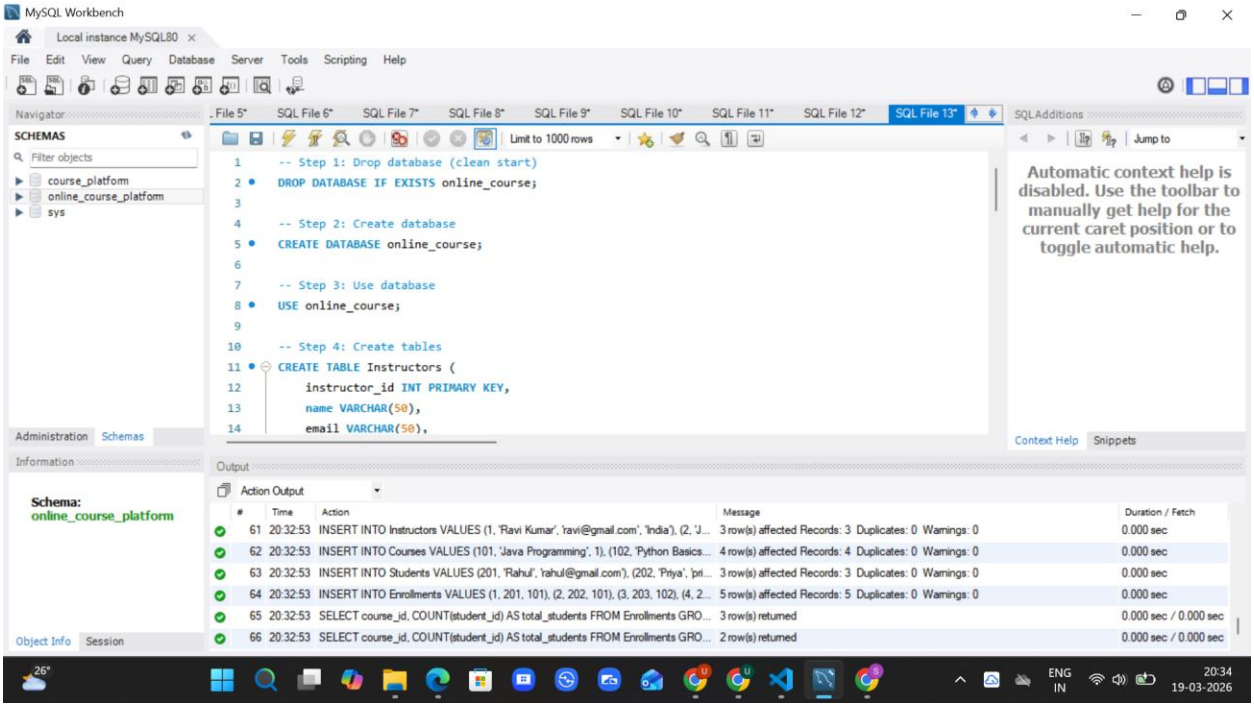
-->Deleted a course using DELETE and verified changes

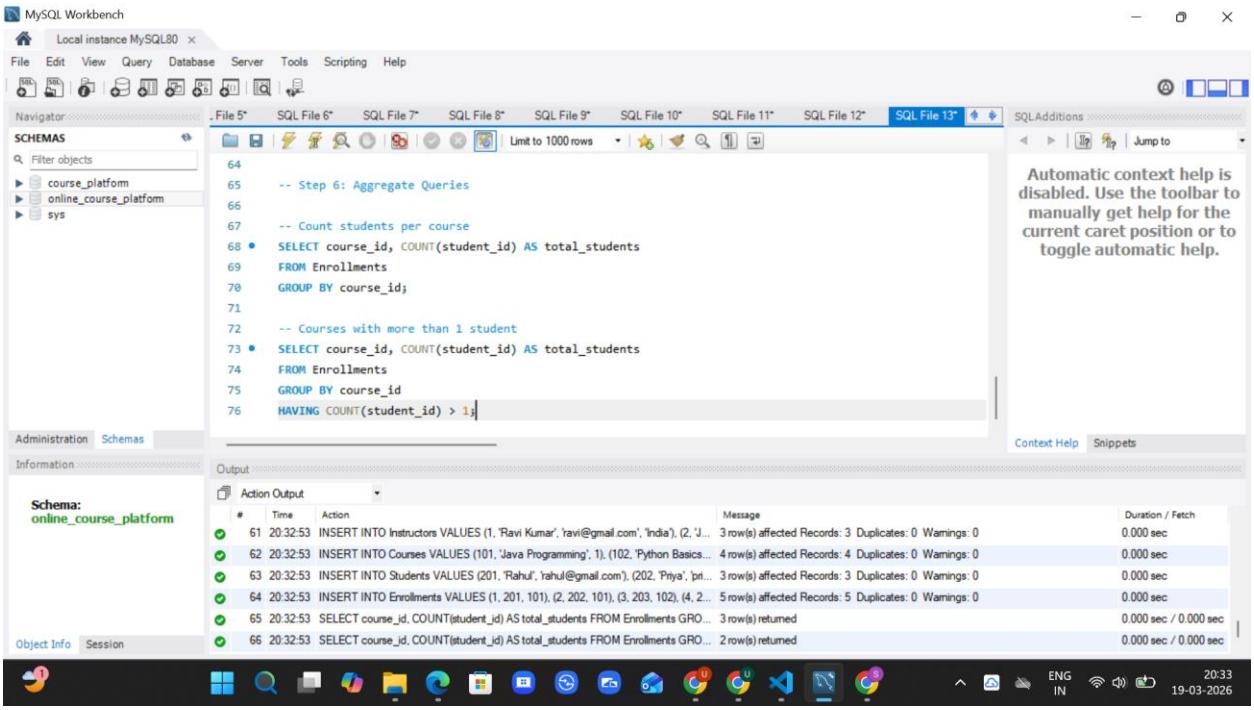
Task 4: Reporting Using Aggregate Queries

Prompt:

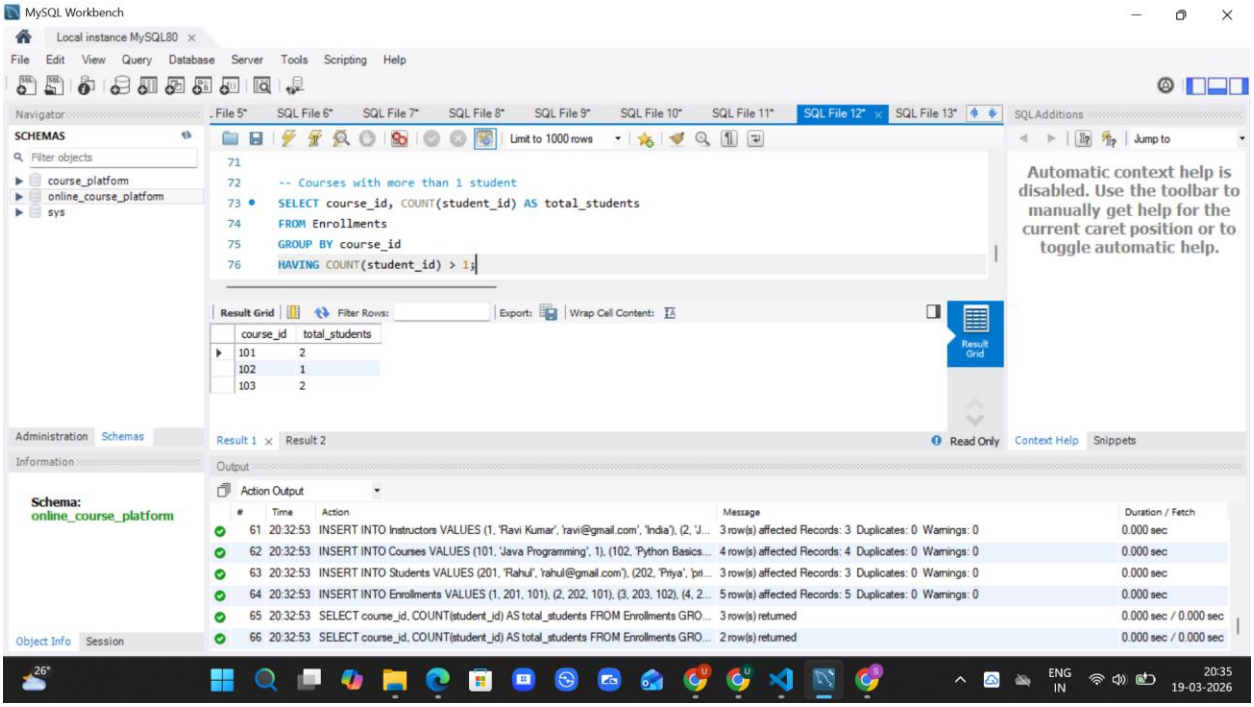
Generate SQL queries to **count students per course** and **list courses with more than a specified number of enrollments** using `GROUP BY` and `HAVING`.

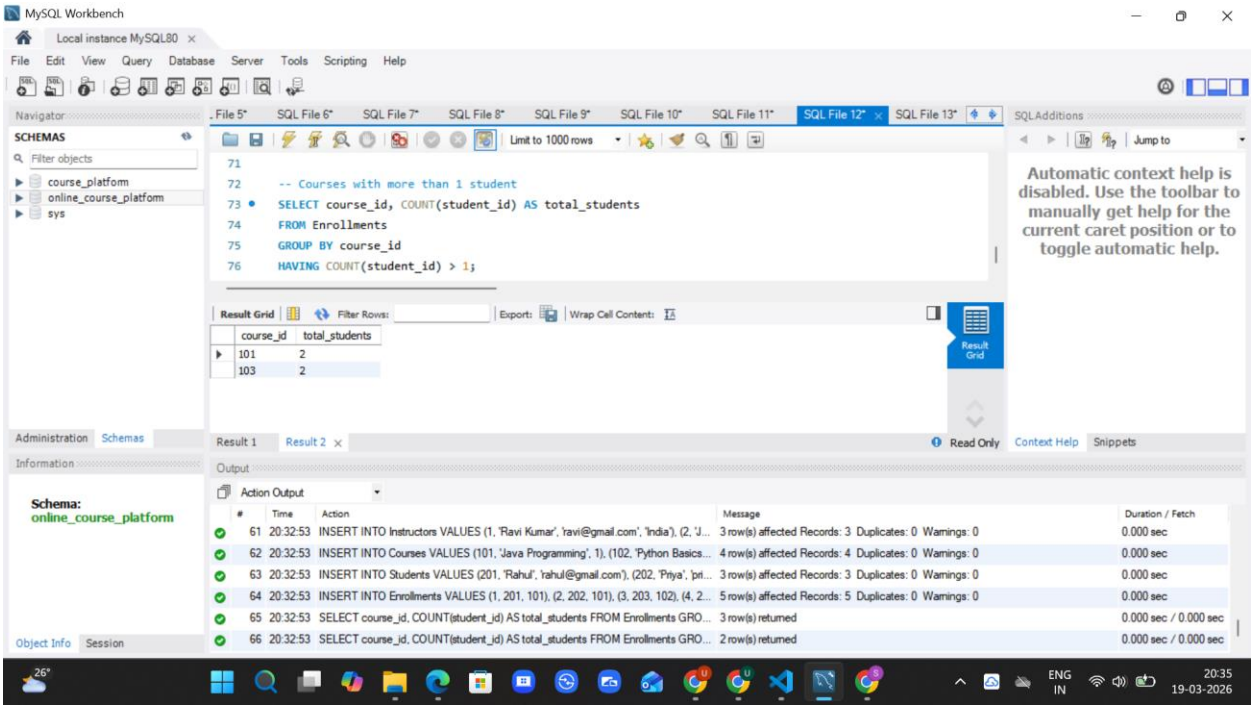
Code:


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*
Limit to 1000 rows
-- Step 1: Drop database (clean start)
DROP DATABASE IF EXISTS online_course;
-- Step 2: Create database
CREATE DATABASE online_course;
-- Step 3: Use database
USE online_course;
-- Step 4: Create tables
CREATE TABLE Instructors (
instructor_id INT PRIMARY KEY,
name VARCHAR(50),
email VARCHAR(50),
Schema: online_course_platform
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Output
Action Output
61 20:32:53 INSERT INTO Instructors VALUES (1, 'Ravi Kumar', 'ravi@gmail.com', 'India'), (2, 'J... 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 0.000 sec
62 20:32:53 INSERT INTO Courses VALUES (101, 'Java Programming', 1), (102, 'Python Basics... 4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0 0.000 sec
63 20:32:53 INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'pri... 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 0.000 sec
64 20:32:53 INSERT INTO Enrollments VALUES (1, 201, 101), (2, 202, 101), (3, 203, 102), (4, 2... 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 0.000 sec
65 20:32:53 SELECT course_id, COUNT(student_id) AS total_students FROM Enrollments GRO... 3 row(s) returned 0.000 sec / 0.000 sec
66 20:32:53 SELECT course_id, COUNT(student_id) AS total_students FROM Enrollments GRO... 2 row(s) returned 0.000 sec / 0.000 sec
20:34 19-03-2026


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*
Limit to 1000 rows
-- Step 6: Aggregate Queries
-- Count students per course
SELECT course_id, COUNT(student_id) AS total_students
FROM Enrollments
GROUP BY course_id;
-- Courses with more than 1 student
SELECT course_id, COUNT(student_id) AS total_students
FROM Enrollments
GROUP BY course_id
HAVING COUNT(student_id) > 1;
Schema: online_course_platform
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Output
Action Output
61 20:32:53 INSERT INTO Instructors VALUES (1, 'Ravi Kumar', 'ravi@gmail.com', 'India'), (2, 'J... 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 0.000 sec
62 20:32:53 INSERT INTO Courses VALUES (101, 'Java Programming', 1), (102, 'Python Basics... 4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0 0.000 sec
63 20:32:53 INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'pri... 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 0.000 sec
64 20:32:53 INSERT INTO Enrollments VALUES (1, 201, 101), (2, 202, 101), (3, 203, 102), (4, 2... 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 0.000 sec
65 20:32:53 SELECT course_id, COUNT(student_id) AS total_students FROM Enrollments GRO... 3 row(s) returned 0.000 sec / 0.000 sec
66 20:32:53 SELECT course_id, COUNT(student_id) AS total_students FROM Enrollments GRO... 2 row(s) returned 0.000 sec / 0.000 sec
20:33 19-03-2026

Output:





Explanation:

-->Used GROUP BY to count students in each course

-->Used COUNT() function for aggregation

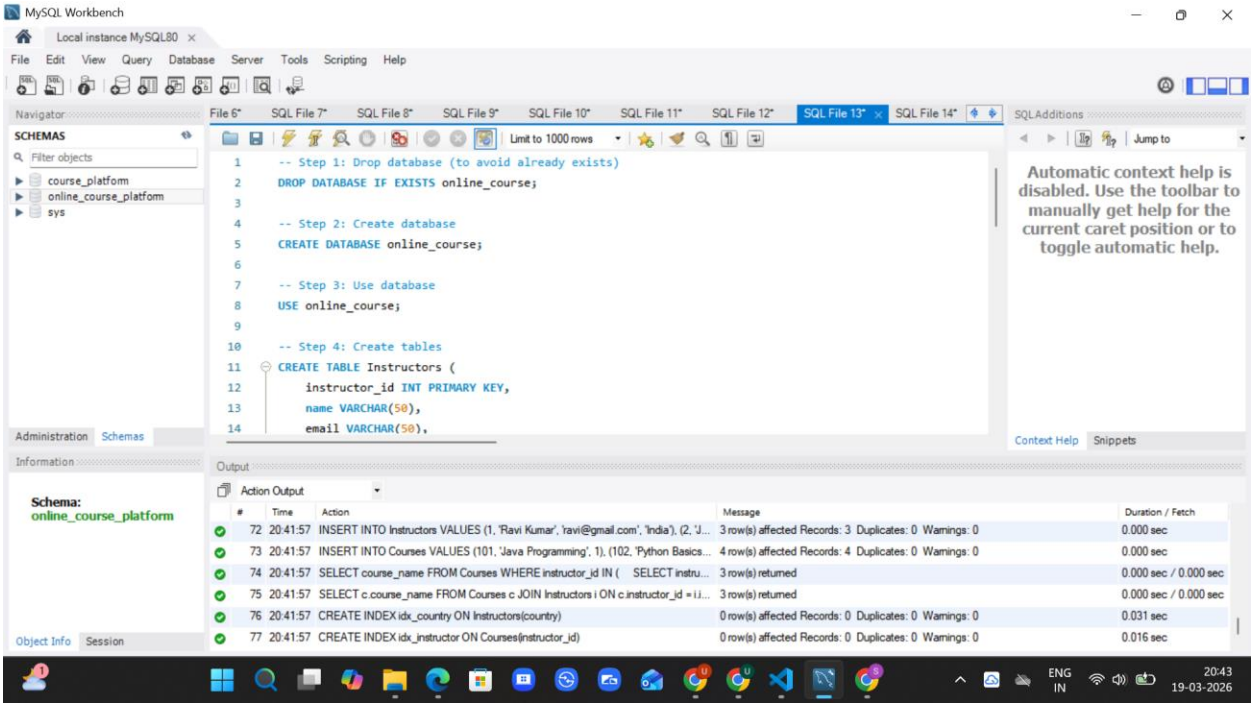
-->Used HAVING to filter courses with more than 1 student

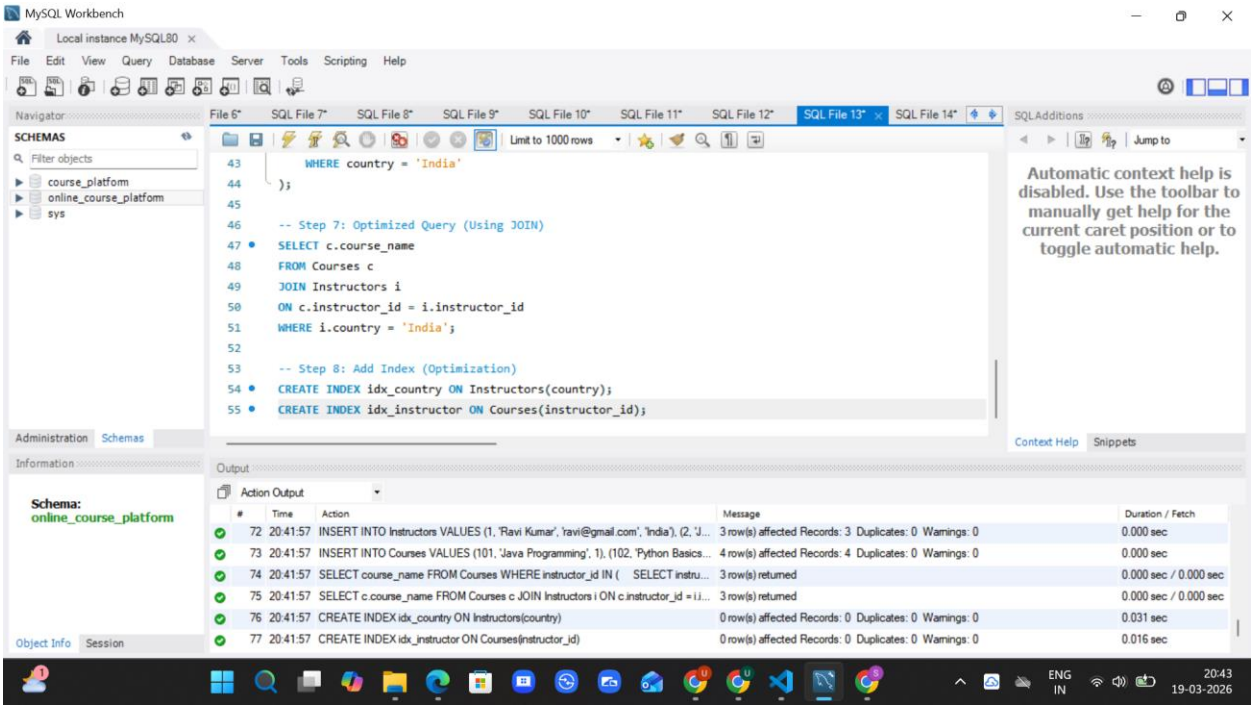
Task 5: Query Optimization with AI Suggestions

Prompt:

Analyze and optimize a query that retrieves courses taught by instructors from a specific country using **efficient joins and indexing**, and compare results.

Code:


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
course_platform
online_course_platform
sys
File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13* SQL File 14*
Limit to 1000 rows
-- Step 1: Drop database (to avoid already exists)
DROP DATABASE IF EXISTS online_course;
-- Step 2: Create database
CREATE DATABASE online_course;
-- Step 3: Use database
USE online_course;
-- Step 4: Create tables
CREATE TABLE Instructors (
instructor_id INT PRIMARY KEY,
name VARCHAR(50),
email VARCHAR(50),
SQL Additions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Administration Schemas
Information
Schema:
online_course_platform
Object Info Session
Output
Action Output
Time Action Message Duration / Fetch
72 20:41:57 INSERT INTO Instructors VALUES (1, 'Ravi Kumar', 'ravi@gmail.com', 'India'), (2, 'J... 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 0.000 sec
73 20:41:57 INSERT INTO Courses VALUES (101, 'Java Programming', 1), (102, 'Python Basics... 4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0 0.000 sec
74 20:41:57 SELECT course_name FROM Courses WHERE instructor_id IN (SELECT instru... 3 row(s) returned 0.000 sec / 0.000 sec
75 20:41:57 SELECT c.course_name FROM Courses c JOIN Instructors i ON c.instructor_id = i.i... 3 row(s) returned 0.000 sec / 0.000 sec
76 20:41:57 CREATE INDEX idx_country ON Instructors(country) 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.031 sec
77 20:41:57 CREATE INDEX idx_instructor ON Courses(instructor_id) 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.016 sec
ENG IN
20:43
19-03-2026


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
course_platform
online_course_platform
sys
File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13* SQL File 14*
Limit to 1000 rows
WHERE country = 'India'
);
-- Step 7: Optimized Query (Using JOIN)
SELECT c.course_name
FROM Courses c
JOIN Instructors i
ON c.instructor_id = i.instructor_id
WHERE i.country = 'India';
-- Step 8: Add Index (Optimization)
CREATE INDEX idx_country ON Instructors(country);
CREATE INDEX idx_instructor ON Courses(instructor_id);
SQL Additions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Administration Schemas
Information
Schema:
online_course_platform
Object Info Session
Output
Action Output
Time Action Message Duration / Fetch
72 20:41:57 INSERT INTO Instructors VALUES (1, 'Ravi Kumar', 'ravi@gmail.com', 'India'), (2, 'J... 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 0.000 sec
73 20:41:57 INSERT INTO Courses VALUES (101, 'Java Programming', 1), (102, 'Python Basics... 4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0 0.000 sec
74 20:41:57 SELECT course_name FROM Courses WHERE instructor_id IN (SELECT instru... 3 row(s) returned 0.000 sec / 0.000 sec
75 20:41:57 SELECT c.course_name FROM Courses c JOIN Instructors i ON c.instructor_id = i.i... 3 row(s) returned 0.000 sec / 0.000 sec
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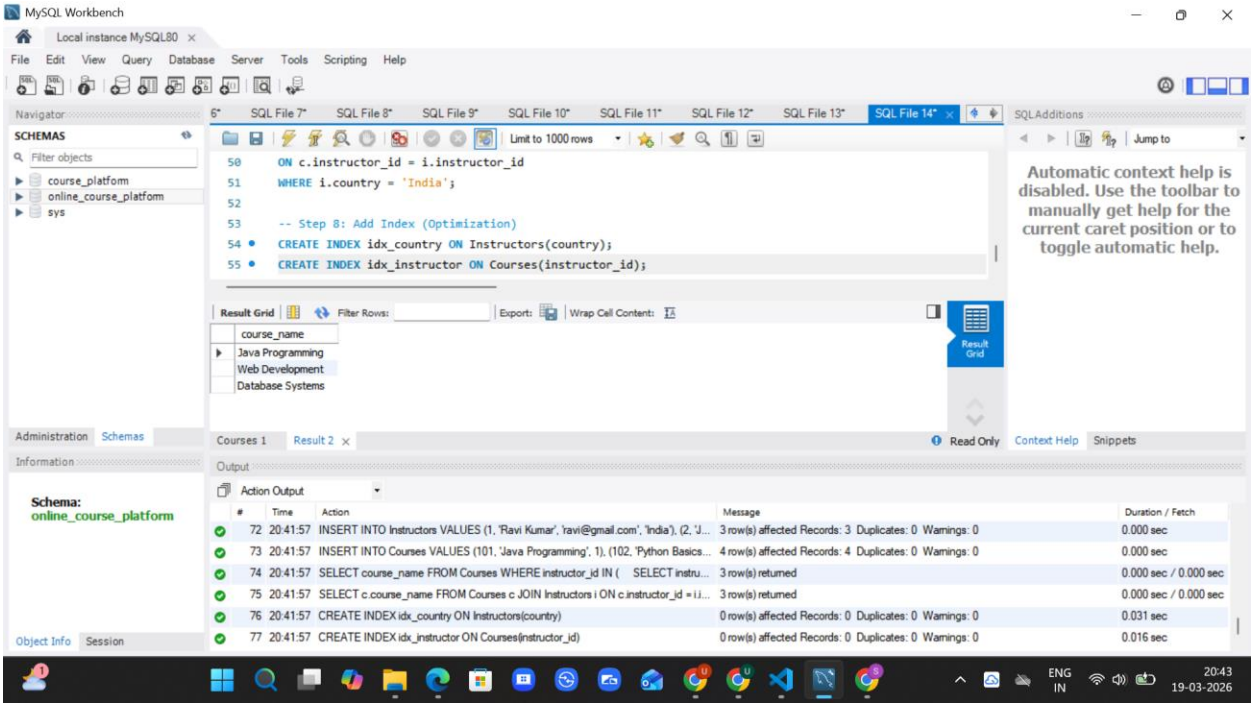
77 20:41:57 CREATE INDEX idx_instructor ON Courses(instructor_id) 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.016 sec
ENG IN
20:43
19-03-2026

Output:



Explanation:

-->Used a normal query with subquery to fetch courses by instructor country

-->Optimized using JOIN for better performance and readability

-->Added indexing to improve query speed on large datasets